Healthcare Revenue and Performance Analytics

Tools & Technologies: Power BI | SQL | DAX | Power Query | Excel

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Healthcare Revenue and Performance Analytics

Detailed Report on Financial Performance, Collections, Insurance, and Cost Trends

Prepared from Power BI Dashboard Analysis, July 2024 to August 2026

1. Project Overview

A data analytics project focused on analyzing healthcare financial and operational performance to provide insights into revenue, collections, outstanding balances, departmental performance, profitability, budget performance, and potential revenue leakage.

2. Executive Summary

Healthcare organizations generate large volumes of financial and operational data across billing, payments, departmental activities, patient services, and costs. Transforming this data into meaningful insights is essential for understanding financial performance, improving revenue realization, managing outstanding balances, and supporting effective decision-making.

The **Healthcare Revenue and Performance Analytics** project was developed to provide a consolidated analytical view of key healthcare financial and operational performance indicators. The analysis focuses on areas including revenue performance, payment collections, outstanding balances, departmental contribution, profitability, budget performance, and potential revenue leakage.

An end-to-end analytics approach was followed, beginning with data preparation and validation, followed by data modeling and the development of business-focused KPIs. Interactive dashboards were created in Microsoft Power BI to enable stakeholders to monitor performance, identify trends and variances, and explore areas requiring further attention.

The project demonstrates how healthcare data can be transformed into actionable business intelligence. By bringing together financial and operational perspectives, the dashboards provide a foundation for monitoring performance and supporting data-driven decisions related to revenue optimization, collection management, cost awareness, and overall financial performance.

This documentation presents the complete analytical journey, including the business objectives, data preparation process, data model, KPI framework, dashboard design, key findings, recommendations, and technical implementation.

3. Business Problem & Project Objectives

3.1 Business Problem

Healthcare organizations manage financial information across multiple business processes, including billing, payments, departmental operations, insurance, admissions, and operating costs.

When these areas are analyzed independently, it can become difficult for management to obtain a consolidated view of financial performance and identify areas that require attention.

The project addresses the need for a centralized analytical view of healthcare revenue and financial performance. The analysis focuses on understanding how revenue is generated, how effectively it is converted into collections, where outstanding balances exist, how departments contribute to overall revenue, and where potential revenue leakages or financial performance gaps may occur.

Another key challenge is the need to compare financial performance across different periods and against planned or expected performance. Monitoring trends, variances, and operational indicators can help management identify emerging issues and prioritize areas for further investigation.

The objective of the analysis is therefore not simply to report financial figures, but to transform underlying healthcare data into meaningful business insights that can support revenue monitoring, collection management, financial planning, and performance evaluation.

3.2 Project Objectives

The primary objectives of the Healthcare Revenue Intelligence project are:

1. Analyze Revenue Performance
Evaluate overall revenue performance and identify trends across time, departments, and other relevant business dimensions.

2. Monitor Collection Performance
Assess the relationship between billed revenue and collected amounts, while identifying outstanding balances and potential collection gaps.

3. Identify Potential Revenue Leakage
Analyze financial discrepancies and performance gaps that may indicate potential revenue leakage or opportunities for improved revenue realization.

4. Evaluate Department Performance
Compare revenue contribution and financial performance across departments to identify high-performing and lower-performing areas.

5. Assess Profitability and Cost Performance
Examine the relationship between revenue and operating costs to understand overall financial efficiency and profitability.

6. Compare Budget and Actual Performance
Evaluate actual financial performance against budgeted or planned values and identify significant variances requiring management attention.

7. Enable Management-Level Monitoring
Provide interactive dashboards and KPIs that allow stakeholders to monitor financial performance, investigate trends, and identify areas requiring further analysis.

3.3 Key Business Questions

The analysis was designed to answer the following key business questions:

How is overall healthcare revenue performing over time?

Which departments contribute the most to total revenue?

How much revenue has been collected versus what remains outstanding?

How efficiently is revenue being converted into cash collections?

Where are potential revenue leakage opportunities?

How do operating costs affect profitability?

How does actual performance compare with budget?

Which financial or operational areas require management attention?

What trends or variances should stakeholders investigate further?

3.4 Intended Business Value

The analysis provides a consolidated view of financial performance that can help stakeholders:

Monitor revenue and collection trends.

Identify outstanding balances and collection gaps.

Prioritize potential revenue leakage areas for investigation.

Understand departmental financial contribution.

Monitor profitability and cost performance.

Track budget variances.

Support more informed, data-driven financial decisions.

The dashboards are intended to function as an analytical monitoring and decision-support tool rather than replace detailed financial reconciliation or operational investigation.

4. Dataset Overview

4.1 Data Model Summary

The project uses a structured analytical data model designed to support healthcare revenue, financial performance, collection, cost, insurance, and revenue leakage analysis.

The final reporting model consists of **six dimension tables and seven fact tables**. Dimension tables provide descriptive attributes used to categorize and filter the transactional data, while fact tables contain the measurable financial and operational events used for analysis.

The model covers the following analytical domains:

- Patient and admission activity
- Revenue generation
- Payments and collections
- Insurance claims
- Departmental costs
- Budget versus actual performance
- Revenue leakage

Raw source tables used during the development process are excluded from the final reporting model.

4.2 Dimension Tables

Dimension Table	Records	Analytical Purpose
Dim_Department	15	Department classification and departmental analysis
Dim_Doctor	150	Doctor-level analysis
Dim_Insurance_Provider	50	Insurance provider classification
Dim_Patient	129,815	Patient-level analysis
Dim_Payment_Method	8	Payment method classification
Dim_Revenue_Category	11	Revenue service/category classification

The dimension tables provide the descriptive context required to analyze the fact tables by department, doctor, patient, insurance provider, payment method, and revenue category.

4.3 Fact Tables

Fact Table	Records	Analytical Purpose
Fact_Admissions	300,000	Patient admission and discharge activity
Fact_Budget	375	Budget and actual financial performance
Fact_Department_Cost	375	Departmental cost tracking
Fact_Insurance_Claim	50,000	Insurance claim and settlement analysis
Fact_Payment	250,000	Payment and collection analysis
Fact_Revenue_Leakage	30,000	Potential revenue leakage analysis
Fact_Revenue_Transaction	300,000	Revenue transaction and financial performance analysis

The fact tables represent the primary analytical events and financial measures used throughout the Power BI reporting model.

4.4 Data Coverage

The dataset covers approximately three years, although individual business processes have different date ranges.

Business Area	Start	End
Revenue Transactions	21-Jul-2024	21-Jul-2026
Admissions	21-Jul-2024	21-Jul-2026

Payments	21-Jul-2024	18-Oct-2026
Budget	Jul-2024	Jul-2026
Department Costs	Jul-2024	Jul-2026

The difference in end dates is relevant when comparing revenue, payment, and collection performance across periods.

4.5 Dataset Scale

The analytical model provides substantial operational and financial coverage:

Measure	Volume
Unique Patients	129,815
Admissions	300,000
Revenue Transactions	300,000
Unique Bills	189,678
Payment Records	250,000
Insurance Claims	50,000
Revenue Leakage Records	30,000
Departments	15
Doctors	150
Insurance Providers	50
Payment Methods	8
Revenue Categories	11

The model therefore supports analysis across both **large-scale transactional activity** and **descriptive business dimensions**.

4.6 Analytical Granularity

The dataset operates at multiple levels of granularity.

The **revenue transaction table** contains 300,000 transactions associated with 189,678 unique bills. This means a single bill can contain multiple revenue transactions.

The average number of revenue transactions per bill is **1.58**, with a maximum of **8 transactions for a single bill**.

Therefore:

- **Transaction-level analysis** is appropriate for detailed revenue analysis.
- **Bill-level analysis** requires aggregation of associated transactions.
- **Patient-level analysis** requires aggregation across the patient's transactions or admissions.
- **Monthly analysis** is supported by transaction dates and monthly budget/cost records.

5. Data Model

5.1 Modeling Approach

The Power BI model follows a dimensional analytical approach, bringing together healthcare financial and operational data into a unified reporting model.

The model uses shared dimensions to provide consistent filtering and analysis across multiple business processes, while transactional fact tables provide the underlying measures for financial and operational reporting.

5.2 Relationship Design

Relationships are established using common business keys between the analytical tables.

The primary analytical paths connect:

- Revenue with patients, departments, doctors, and revenue categories
- Admissions with patients
- Payments with payment methods
- Insurance claims with insurance providers
- Budget and departmental costs with departments
- Transactional financial data through common business references

This structure enables users to analyze financial performance across different organizational and operational perspectives without requiring separate reporting models for each business area.

5.3 Date Modeling

A centralized Dim_Date table provides the time dimension for the reporting model.

Different business processes contain their own relevant dates, including transaction, payment, admission, claim submission, leakage, budget, and cost dates.

The model uses the appropriate date context for each analysis, supporting:

- Monthly trends
- Year-over-year comparisons
- Month-over-month analysis
- Budget-period comparisons
- Period-based financial reporting

5.4 Analytical Granularity

The model supports analysis across multiple levels, from detailed transactions to aggregated business performance.

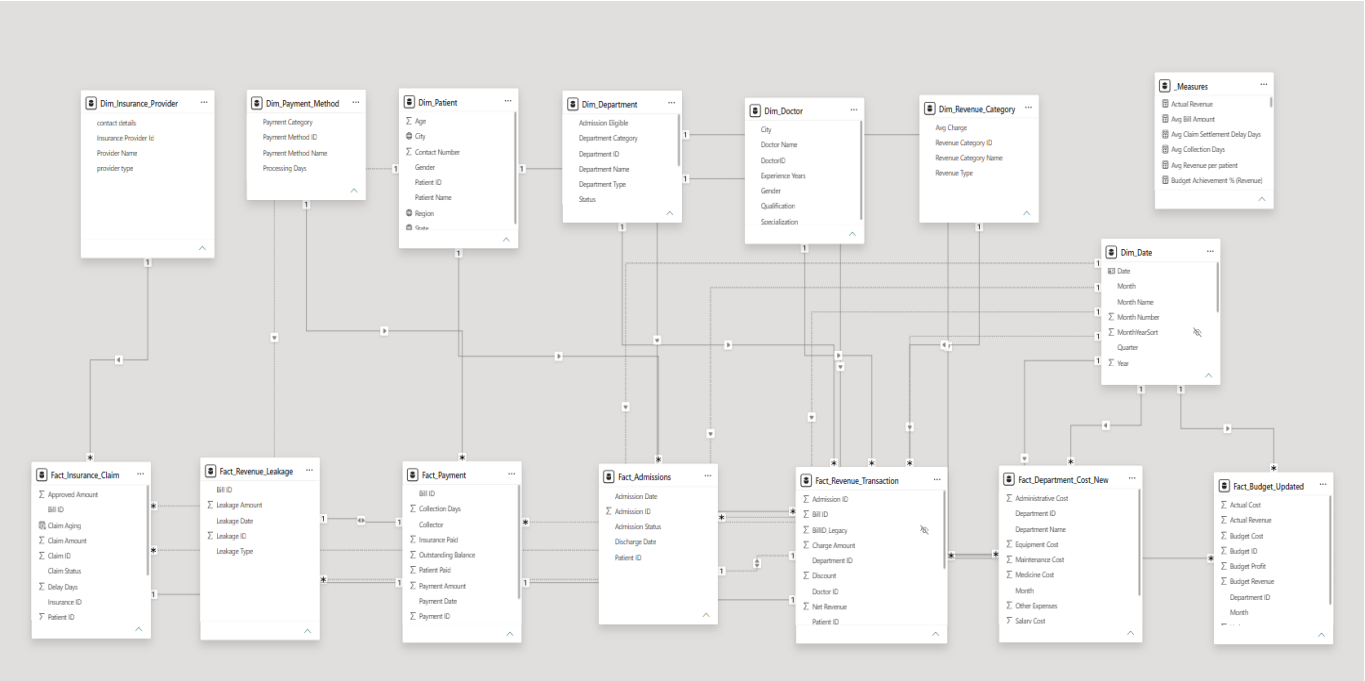
Revenue is recorded at the **transaction level**, while other processes such as payments, claims, admissions, budget, and departmental costs operate at their respective business-event or periodic levels.

This allows the reporting layer to provide both:

Detailed analysis → transaction, bill, patient, or event level

Aggregated analysis → department, revenue category, monthly, and yearly level

The model therefore provides a foundation for moving from **executive-level KPIs to detailed business drivers**.



5.5 Role of the Data Model in Reporting

The data model serves as the foundation for the Power BI reporting layer.

It enables:

- Consistent filtering across dashboards
- Cross-functional financial analysis
- Time-based performance analysis
- Drill-down from summary KPIs to underlying business dimensions
- Integration of revenue, cost, payment, insurance, and leakage perspectives

The model was designed with the reporting requirements in mind so that the dashboards can present a unified view of **healthcare financial performance and revenue management**.

6. Analytical Framework

The analysis evaluates hospital financial performance across six core areas: **revenue, budget and cost, collections, insurance claims, revenue leakage, and departmental/service performance**, supported by time-based analysis.

Analytical Area	Key Focus
Revenue Performance	Revenue trends, department and category contribution, charges, discounts, tax, and net revenue
Budget & Cost Performance	Budget vs. actual revenue/cost, variance, profitability, and cost composition
Collections & Payment Performance	Collections, outstanding balances, payment status, collection days, and payment components
Insurance Claims Performance	Claim amounts, approvals, rejections, settlements, status, delays, and provider performance
Revenue Leakage	Total leakage, leakage types, affected bills, trends, and category contribution
Department & Service Analysis	Department revenue, service-category performance, costs, and financial contribution
Time-Based Analysis	Monthly/yearly trends, MoM, YoY, YTD, and budget-period comparisons

The framework brings these perspectives together to provide a unified view of financial performance and the key factors influencing revenue realization and profitability.

7. Power BI Implementation

The analytical framework was implemented in **Power BI** as an interactive financial intelligence report. The reporting layer combines KPIs, trend analysis, comparative visuals, interactive filters, and detailed breakdowns to support both executive monitoring and focused financial analysis.

7.1 Data & Model Integration

- Connected the structured analytical tables to Power BI.
- Applied the established data model and relationships.
- Used a centralized date dimension for time-based analysis.

7.2 Data Transformation & Measures

- Performed required data preparation using **Power Query**.
- Developed **DAX measures** for financial KPIs, comparisons, trends, and performance indicators.
- Implemented time-intelligence calculations such as **MoM, YoY, and YTD** where applicable.

7.3 Visual Analytics

The report uses a combination of:

- KPI cards
- Trend and area charts
- Bar and column charts
- Donut/Pie charts
- treemap visual
- Waterfall charts
- Ribbon charts
- Matrix-based analysis
- Interactive tooltips

Visuals were selected based on the analytical question rather than using a fixed visualization format.

7.4 Interactivity & Navigation

Interactive features allow users to explore the analysis by relevant dimensions such as:

- Year and time period
- Department
- Region
- State
- Other applicable business dimensions

Slicers, cross-filtering, drill-downs, and report-level interactions enable users to move from summary-level performance to more detailed analysis.

7.5 Executive Reporting

The report is designed to provide a high-level view of financial performance while allowing users to investigate the underlying drivers of:

Revenue → Collections → Insurance → Leakage → Cost → Profitability

This structure supports both **performance monitoring and business-oriented investigation**.

8. Dashboard Overview

The Power BI report consists of nine interactive dashboards, each focused on a specific aspect of hospital financial performance.

Dashboard Page	Purpose
Executive Financial Overview	Provides a consolidated view of key financial KPIs and overall business performance for executive-level monitoring.
Financial Analytics	Provides detailed analysis of revenue and financial performance across relevant business dimensions.
Insurance Analytics	Evaluates insurance claims, approvals, rejections, settlements, and claim-related performance.

Collection Performance	Monitors collections, outstanding balances, payment status, and collection efficiency.
Departmental Financial Performance	Compares financial performance across departments and identifies major departmental contributors.
Revenue Leakage	Identifies and analyzes revenue leakage, including leakage types, affected bills, and financial impact.
Budget vs Actuals	Compares planned and actual revenue and costs to identify financial variances and performance gaps.
Cost Optimization	Analyzes departmental cost components to identify major expense areas and potential cost-control opportunities.
Cash Flow & Financial Trends	Tracks financial trends and cash-related performance over time to support monitoring of revenue realization and financial movement.

Dashboard Interactivity

Across the report, interactive slicers, filters, drill-downs, and tooltips allow users to explore financial performance by relevant time periods, departments, regions, states, revenue categories, and other applicable dimensions.

The dashboards are designed to progress from **executive-level monitoring to focused financial investigation**, enabling users to identify performance changes and investigate their underlying drivers.

9. Key Business Insights

The analysis identified several key financial patterns across revenue generation, profitability, collections, insurance, leakage, and cost performance.

Area	Key Insight
Revenue	Total net revenue reached **₹511.05M**, with a **71.79% revenue-to-cost margin**.
Revenue Concentration	**Surgery** generated **56.82%** of total net revenue, followed by Operation Theatre (**14.22%**) and ICU (**11.21%**).
Department Performance	**Pathology, Pharmacy, and Emergency** were the leading departments by net revenue.
Budget Performance	Actual revenue was **₹511.05M** against a budget of **₹514.15M**, resulting in an overall revenue variance of approximately **-₹3.10M**.
Cost Performance	Actual cost was **₹144.16M**, slightly below the budgeted **₹144.30M**, indicating a favorable cost variance of approximately **₹0.14M**.
Collections	Total payments amounted to **₹213.33M**, while **₹42.66M** remained outstanding.
Collection Status	**249,972 payment records** were classified as partial, compared with only **28 completed** payments.

Insurance Claims	Claims totaled **₹58.62M**, of which **₹51.32M** was approved and **₹7.29M** was rejected.
Claim Processing	Average claim delay was approximately **95 days**, highlighting a significant processing/settlement cycle.
Revenue Leakage	Identified leakage totaled **₹9.05M** across 30,000 records.
Major Leakage Source	**Discount Leakage** was the largest category at approximately **₹3.23M**, followed by Insurance Rejection at **₹2.75M**.
Cost Structure	**Salary Cost (₹70.89M)** was the largest departmental cost component, followed by Medicine and Equipment costs.

Revenue & Business Performance

Strong revenue with moderated growth:
Total net revenue reached **₹511.05M**. Revenue increased from **₹115.1M in 2024 to ₹254.0M in 2025**, while 2026 reached **₹141.9M through July**. The post-2025 trend indicates a moderation in growth.

High revenue concentration in Surgery:
Surgery contributed **₹290.4M, or 56.8% of total revenue**, making it the dominant revenue category. Its contribution remained relatively consistent across regions.

Regional contribution:
The South region generated **₹184.3M, representing 36% of total revenue**. Its higher absolute claims and outstanding balances are broadly proportional to its patient and revenue contribution rather than indicating a clear regional efficiency gap.

Collections & Insurance

Collection cycle is lengthening:
Average collection days increased consistently from **39.0 days in 2024 to 44.6 days in 2025 and 47.2 days in 2026**, indicating a persistent deterioration in collection speed.

Insurance settlement remains a bottleneck:
Insurance claims totaled **₹58.62M**, with an **87.6% approval rate** and **12.4% rejection rate**. Average settlement delay remained around **95 days** across the three-year period, indicating a persistent rather than worsening processing bottleneck.

Profitability & Cost

Strong margin with improving cost efficiency:
Total operational cost was **₹144.16M**, representing **28.2% of revenue**, resulting in a **71.8% margin**. The cost-to-revenue ratio improved from **29.9% in 2024 to 27.0% in 2026**.

Nephrology and Oncology require attention:
These departments combine relatively high costs with lower margins, making them potential areas for deeper cost and profitability review.

Revenue Leakage

Leakage is increasing:
Identified revenue leakage totaled **₹9.05M**, approximately **1.8% of net revenue**. The leakage rate increased from **1.12% in 2024 to 1.79% in 2025 and 2.26% in 2026**, indicating a growing financial risk.

Discount and insurance-related leakage are the largest contributors:
Discount leakage and insurance rejection represent the largest leakage categories, followed by cancelled bills and coding errors.

Budget & Cash Performance

Budget execution remains strong:
Actual revenue reached **₹511.05M against a ₹514.15M budget**, achieving **99.4%** of the revenue target. Actual cost was approximately **₹0.14M below budget**.

Cash realization is improving:
Net cash flow reached **₹69.17M**, while cash realization improved from **30.3% in 2024 to 41.7% in 2025 and 51.1% in 2026**. This represents one of the strongest positive trends identified in the analysis.

10. Recommendations

The findings suggest several areas where management attention can improve revenue realization, financial efficiency, and operational control.

1. Address the Rising Collection Cycle

Average collection days have increased for three consecutive years. Strengthening follow-up processes and monitoring aging and outstanding balances can help accelerate cash realization.

2. Reduce Revenue Leakage

The increasing leakage rate requires closer monitoring, particularly for **discount leakage and insurance-related rejections**. Establishing regular leakage reviews can help identify recurring causes and reduce preventable losses.

3. Improve Insurance Claim Processing

The approximately **95-day settlement cycle** indicates a persistent bottleneck. Reviewing delayed claims, rejection patterns, and provider-level performance can help improve settlement efficiency.

4. Review Departmental Cost Efficiency

Nephrology and Oncology should receive focused profitability reviews because of their combination of relatively high costs and lower margins. Cost drivers should be evaluated while maintaining required service quality.

5. Maintain Budget Discipline

Overall budget achievement remains close to target, but departments with unfavorable variances, particularly **Emergency**, should be monitored through regular budget-versus-actual reviews.

6. Monitor Revenue Concentration

Surgery contributes more than half of total revenue. Its strong performance should be maintained while management evaluates opportunities to strengthen other revenue-generating services and reduce excessive dependence on a single category.

7. Sustain Improvements in Cash Realization

The improvement in cash realization is a positive trend. The practices contributing to this improvement should be maintained while simultaneously addressing the rising collection cycle.

Overall Recommendation

The analysis indicates that the primary opportunity is to **strengthen revenue realization while preserving the organization's strong profitability and cost efficiency**. Priority should be given to collection-cycle improvement, leakage reduction, insurance processing, and targeted departmental cost management.

11. Limitations & Assumptions

The analysis and reporting were developed within the scope of the available dataset and reporting period.

- The analysis covers financial and operational data for **2024–2026**, with 2026 representing **year-to-date performance through July** for applicable metrics.
- Different tables may have different date coverage based on the underlying business process.

- Revenue, payment, claim, leakage, budget, and cost metrics are interpreted according to their defined business logic within the analytical model.
- Budget-versus-actual comparisons are evaluated for corresponding reporting periods.
- Dashboard results are dependent on the completeness and quality of the underlying data.
- The findings are intended for **analytical assessment and decision support** and should be validated with relevant operational and financial teams before business decisions are implemented.

12. Conclusion

The Hospital Financial Performance & Revenue Intelligence project provides an integrated view of hospital through interactive Power BI dashboards.

The analysis brings together revenue, budget, cost, collections, insurance claims, revenue leakage, and cash realization to identify key financial drivers and areas requiring attention.

The findings indicate strong financial performance revenue generation, profitability, improving cost efficiency, and positive cash-realization trends**,** while also highlighting opportunities to improve collection cycles, insurance settlement efficiency, revenue leakage, and departmental cost performance**.**

The resulting Power BI reporting framework enables stakeholders to move from high-level financial monitoring to focused financial investigation**,** supporting data-driven decision-making and more effective financial performance management.